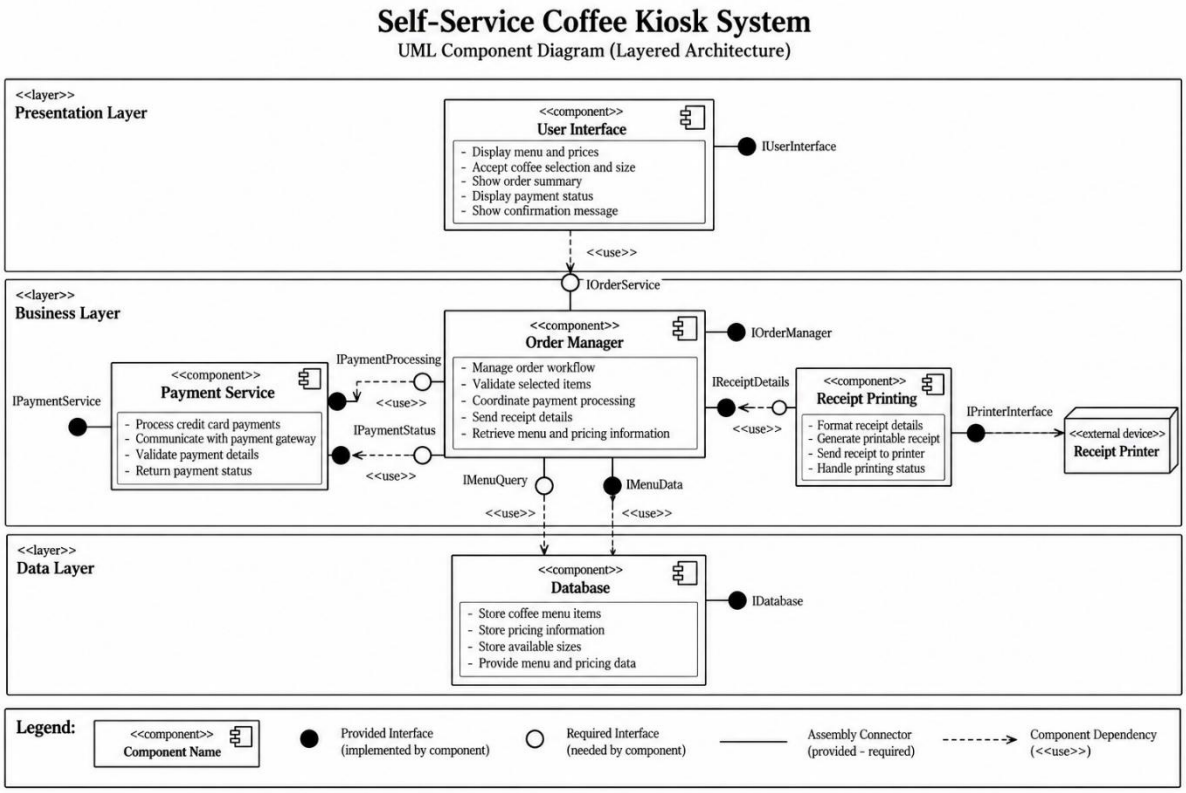
SE LAB 03

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Architecture Selection and Justification

Architecture Selection: We chose Layered Architecture for the Self-Service Coffee Kiosk System.

Reasons for Selection:

1. Separation of Concerns: Layered Architecture divides the system into distinct layers, namely the Presentation Layer, Business Layer, and Data Layer. Each layer is responsible for a specific set of operations. The Presentation Layer handles the touch-screen interface through which customers select their coffee type, choose a drink size, and view their order details. The Business Layer manages order processing, credit card payment requests, and receipt generation. The Data Layer stores and retrieves menu items and pricing information. This separation makes the

system easier to understand, develop, test, maintain, and modify without unnecessarily affecting other parts of the system.

2. Organized Functionality and Maintainability: The architecture assigns specific responsibilities to individual components within the layers. The User Interface Component manages customer interactions, the Order Manager Component coordinates the order workflow, the Payment Service Component handles credit card payments, the Receipt Printing Component prepares and sends receipt details to the printer, and the Database Component manages menu and pricing data. This modular organization makes it easier to identify errors, update menu information, and modify individual components while keeping the overall system structure organized.

3. Security Advantage: Layered Architecture supports better control over access to system functionality by separating customer interactions from business operations and data storage. Payment processing can be handled through the Payment Service Component rather than directly by the user interface, while database access can be managed through the appropriate components. This separation helps reduce unnecessary direct access to sensitive operations and supports a more controlled interaction between the different parts of the system.

4. Performance Benefit: Separating presentation, business logic, and data management allows each component to perform its designated responsibilities without mixing unrelated operations. The User Interface Component can handle customer selections while the Order Manager coordinates order processing and retrieves the required menu and pricing information from the Database Component. This organized flow of operations can reduce unnecessary processing and make the system easier to optimize for quick customer interactions in a busy café.